Quantum Cryptography and The BB84 Protocol

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The idea of secure communication has been around since the beginning of time and finds itself at risk especially now, in the flourishing Information Age when the internet lays the groundwork for most of our intercommunication.

Quantum Cryptography is today considered to be "the science of exploiting quantum mechanical properties to perform cryptographic tasks". In other words, it is the process used for encrypting a message in a way that is only known to the parties involved, by making use of the principles of quantum mechanics.

1 Mathematical Preliminaries

A vector space V is a set where we can define both the addition of two vectors and the multiplication by a scalar, which are in fact the underlying properties describing linear combinations. Also, on a vector space V, there can be defined a vector space of linear functions, called a dual vector space V^*. A Hilbert space $\mathcal{H}$ is a vector space with inner product.

In basic Dirac notation, a *ket* vector is an element of the vector space V and is denoted $|x\rangle$, while a *bra* is a function acting on the vector space and is denoted $\langle\alpha|$. Their inner product is defined as $\langle\alpha|x\rangle = \sum \alpha^* x$. The norm of a vector is found to be $\| |x\rangle \| = \sqrt{\langle x|x\rangle}$ and therefore a vector having the norm 1 is called normalised.

A set of vectors is said to be linearly independent if none of the vectors from the set can be written as a linear combination of the others. In a space of n dimensions, a set of n linearly independent vectors will completely span the space, thus forming a basis. A basis $\{|e_i\rangle\}$ whose vectors are normalised and are all perpendicular to each other is called orthonormal. This is often denoted $\langle e_i|e_j\rangle = \delta_{ij}$.

A linear operator A satisfies the relation $A(c_1 |x\rangle + c_2 |y\rangle) = c_1 A |x\rangle + c_2 A |y\rangle$, where c_1 and c_2 are scalars and $|x\rangle$ and $|y\rangle$ are vectors. For a given linear operator, its Hermitian conjugate $A^\dagger$ is defined as the transpose of the matrix described by the complex conjugates of the elements of the initial matrix: $A^\dagger_{ij} = A^*_{ji}$. An operator is therefore called Hermitian if $A^\dagger = A$.

The tensor product of two matrices A (of dimension m×n) and B (of dimension p×q) is the following:

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & ... & a_{1n}B \\ a_{21}B & a_{22}B & ... & a_{2n}B \\ ... & ... & ... & ... \\ a_{m1}B & a_{m2}B & ... & a_{mn}B \end{pmatrix}.$$

The results is therefore a matrix of dimensions (mp)×(nq).

2 The Framework of Quantum Mechanics

Quantum Mechanics is founded on certain "postulates" that manage to perfectly illustrate the connection between the physical world and the formalism of this theory. These postulates have different turns of phrase in accordance to the author's choice and may also differ in number. Regardless of their wording, they have never been proven theoretically and therefore are only justified as being true by the empirical fact that they are consistent with all the experimental data that is known so far.

The first postulate associates to any physical system a certain Hilbert space $\mathcal{H}$. A pure state is thereby characterised by a normalised vector $|\psi\rangle$ and for every two physical states of a system $|\psi_1\rangle$ and $|\psi_2\rangle$, their linear combination also represents a possible state of the given system. This is commonly known as the principle of superposition. The concepts of Hilbert space, normalised vectors and linear combination have previously been explained.

In the case of a composite system, the second postulate states that the Hilbert space associated to this new system is the tensor product of the Hilbert spaces of the component physical systems. Thus, for n systems prepared in the states $|\psi_1\rangle, |\psi_2\rangle, ... |\psi_n\rangle$, the joint state of the total system becomes $|\psi_1\rangle \otimes |\psi_2\rangle \otimes ... \otimes |\psi_n\rangle$.

The third postulate refers to the evolution in time of a closed quantum system and claims that the time dependence of a state is governed by the Schrödinger equation:

$$H |\psi(t)\rangle = i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle .$$

In this equation, $\hbar$ is a physical constant known as Planck's constant and H is a fixed Hermitian operator known as the Hamiltonian of the system, which dictates the dynamics of the system.

The remaining postulates can also be stated together as they all refer to the collapse upon measurement of the wave function. For any observable attribute a of a physical system, there exists a corresponding Hermitian operator A acting on the given Hilbert space, and for every operator, there are special states that remain unchanged, called the eigenstates. By making a measurement of our observable a, the only possible result we can obtain is one of the eigenvalues λ_i of the operator A. As the set of eigenvectors $\{|\lambda_i\rangle\}$ always forms an orthonormal basis, we can write our wave function in the form $|\psi\rangle = \sum c_i |\lambda_i\rangle$. After the measurement has yielded a value λ_i, the wave function immediately collapses on the corresponding (normalised) eigenvector $|\lambda_i\rangle$, with a probability given by c_i squared. The coefficient c_i^2 represents the square of the inner product of $|\psi\rangle$ with the eigenstate $|\lambda_i\rangle$, $|c_i|^2 = (\langle\lambda_i|\psi\rangle)^2$.

On the whole, the first postulate sets the arena for quantum mechanics, by specifying how the state of an isolated quantum system is to be described, the second one generalises this to composite systems, postulate 3 explains the dynamics of a closed quantum systems with the help of the Schrödinger equation, whilst the last one tells us how to extract information from our system by describing the collapse upon measurement.

3 Qubits

Qubits, or quantum bits, can more easily be understood in correlation with classical bits. The same way a bit can be either 0 or 1, depending on whether current is passing through it (1) or not (0), qubits are unit vectors that can be in a state $|0\rangle$, $|1\rangle$ or a linear combination of the two: $a|0\rangle + b|1\rangle, a^2 + b^2 = 1$. In Quantum Computing, the two orthonormal basis $|0\rangle$ and $|1\rangle$ are usually represented by $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Although a qubit can take infinitely many states, by measuring it we can only obtain $|0\rangle$ or $|1\rangle$ with different probabilities (a^2 and b^2, respectively), as the 4th postulates dictates, which means we can extract from it the same "amount of information" as from a classical bit.

Qubits are usually used in Quantum Computing and therefore the physical significance of $|0\rangle$ and $|1\rangle$ depends on the process implied. For example, $|0\rangle$ could represent vertically polarised photons and $|1\rangle$ horizontally polarised photons or $|0\rangle$ could be used for a $\pi/4$ polarisation anticlockwise and $|1\rangle$ for a $\pi/4$ polarisation clockwise. In both cases orthonormality has been assured.

4 Quantum Key Distribution and The BB84 Protocol

Cryptography is the process of converting an ordinary message into an intelligible text so that only the parties involved can understand the initial message with the help of a key. The principle of cryptography can also be understood as the idea of encrypting and decrypting a certain message, together with the distribution of the corresponding key between the sender and the receiver. Quantum Key Distribution is an extremely secure way of distributing this key between the two parties by making use of qubits, the two of them having the capacity to detect if someone is eavesdropping and therefore knows their key.

If a certain key is being sent through a classical channel of communication, there is always a possibility for someone to eavesdrop. However, if a quantum channel is employed instead, this problem is solved.

BB84 is a quantum key distribution scheme proposed by Charles Bennett and Gilles Brassard in 1984, being the first quantum cryptography protocol developed at the time which is provably to be secure and is as follows:

Suppose there is a classical channel between Alice and Bob, and also a unidirectional quantum channel that Alice uses to send Bob a sequence of qubits. For an easier illustration of the protocol, qubits are chosen to be polarised photons (vertically, horizontally, $\pi/4$ anticlockwise

and $\pi/4$ clockwise).

Alice employs two distinct coding systems, the vertical and horizontal polarisation which is taken to be 0 ($|\uparrow\rangle = 0$ and $|\rightarrow\rangle = 1$) and the $\pi/4$ anticlockwise and $\pi/4$ clockwise which is taken to be 1 ($|\nearrow\rangle = 0$ and $|\nwarrow\rangle = 1$). The alternation of the two coding systems is chosen completely random by Alice. Therefore, if Alice intends to send Bob the sequence 01101001, the 0 she wants to send can be either $|\uparrow\rangle$ or $|\nearrow\rangle$, while 1 can be either $|\rightarrow\rangle$ or $|\nwarrow\rangle$. We can assume she sends a sequence of $4N$ photons in total.

After Bob receives one qubit, he also has to choose in which basis to measure it and therefore employs the 0 and 1 coding systems randomly and independently of Alice's choice (that he has no knowledge of). In this way, after the measurement, Bob ends up with a series of $4N$ classical bits, but not identical to the ones Alice had sent.

After all the photons are sent, the two of them exchange through the classical channel their sequence of the choices used for the coding systems. In this case, the photons that were measured by Bob in the same basis as the one Alice initially chose will undoubtedly correspond to the bit she wanted to send. However, if the choice of basis is different, i.e. Alice chose the basis 0 whilst Bob measured it in 1 or vice versa, the probability of the classical bit to be the same is only 50%, as imposed by the fourth postulate:

$$|\nearrow\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\rightarrow\rangle),$$

$$|\nwarrow\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle - |\rightarrow\rangle).$$

As only the photons measured in the same basis have a 100% probability to coincide, both Alice and Bob drop the ones measured in different basis and therefore, probabilistically speaking, each ends up with the same sequence of approximately $2N$ bits.

At this point, the two of them can easily detect if their communication was secure by exchanging a series of N bits from the $2N$ they have left. If the bits coincide perfectly, the chances are the channel was secure and they can therefore drop the N bits they sent and use the remaining ones as the key. However, if they do not coincide, it can be implied that a third party, Eve, has definitely eavesdropped.

Explicitly, when Eve wants to measure the $4N$ qubits Alice had sent, she also has to randomly and independently employ one of the basis 0 and 1 and then send $4N$ qubits back to Bob in order to hide her presence. In this way, if Alice sent a vertically polarised photon (the bit 0 in the basis 0) and Eve measured it in the basis 1, there is a 50% possibility she measured it as being 1. Once more, Eve has to randomly and independently choose a basis and she can opt to send Bob a horizontally polarised photon with a probability of 50%, in which case Bob will get the bit 1, as measured in basis 0. In this way, there is a probability of 25% that even when Alice and Bob use the same coding and decoding basis, the bits they obtain are different, assuring the high level security of the protocol.

5 References

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